



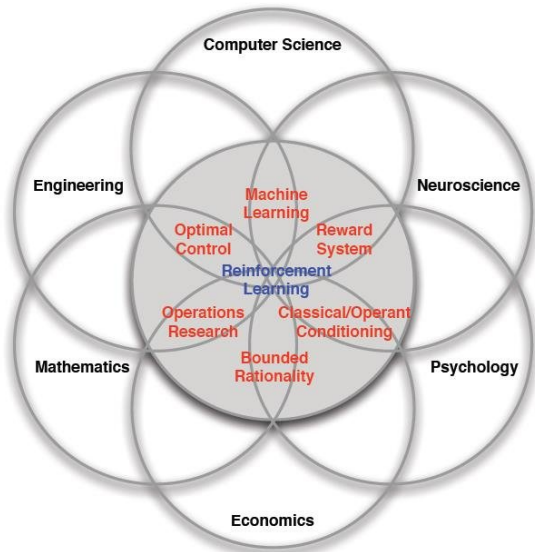
Introduction to Reinforcement Learning

Artificial Intelligence (62651)

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What is reinforcement learning



Credit: David Silver

Reinforcement learning vs (un)supervised machine learning

Reinforcement learning is both a subfield and a problem definition.

- There is an agent that takes **actions**.
- No supervision, labels or oracle. Just a **reward** (good vs bad).
- Dynamic/sequential problem. **Time** is always involved.
- Data and actions are interconnected.
- **Noisy movement**. Non-deterministic dynamics.
- The target is a behaviour, **policy** or controller.
- **Optimal** policy: maximize the reward.

Examples of reinforcement learning

- Fly stunt manoeuvres in a helicopter
 - Defeat the world champion at Go
 - Manage an investment portfolio
 - Control a power station
 - Make a humanoid robot walk
 - Autonomous driving and parking
 - Play many videogames better than humans
 - Design realistic computer simulations.
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- More applications (Csaba Szepesvari live slides)
 - Course introduction slides



Videos

What is a reward?

Reward hypothesis [Sutton and Barto]

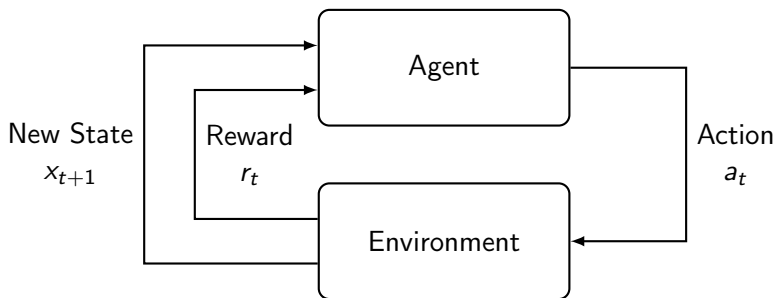
That all of what we mean by **goals** and purposes can be well thought of as the **maximization of the expected value** of the cumulative sum of a received scalar signal (called **reward**).

- A **reward** R_t is a scalar feedback signal (e.g.: score in a game)
- Indicates how well agent is doing at step t
- Objective: maximize cumulative (past, present and future) reward
- In reinforcement learning, people usually work with **rewards**
- In control theory/robotics/engineering, people usually work with **costs**
- *Maximize reward vs minimize cost* $\rightarrow$ reward = -cost

Examples of rewards/costs

- Fly stunt manoeuvres in a helicopter/Autonomous driving and parking
 - ▶ + reward for following desired trajectory
 - ▶ - reward for crashing/energy/fuel
- Defeat the world champion at Go
 - ▶ +/- reward for winning/losing a game
- Manage an investment portfolio
 - ▶ + reward for each \$ in bank
- Control a power station
 - ▶ + reward for producing power
 - ▶ - reward for exceeding safety thresholds
- Make a humanoid robot walk
 - ▶ + reward for forward motion/smoothness
 - ▶ - reward for falling over
- Play many videogames better than humans
 - ▶ +/- reward for increasing/decreasing score
- Design realistic computer simulations.
 - ▶ +/- reward for similarity to real videos.

Markov Decision Processes

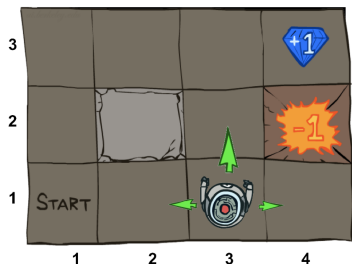


For a [Markov Decision Processes \(MDPs\)](#):

- Markov property $x_t = f(x_{t-1}, a_t)$ (State as memory)
- Action comes from policy and current state $a_t \leftarrow \pi(x_t)$

Example. Gridworld

- Discrete problem (maze-like):
 - ▶ The agent lives in a grid.
Discrete states = grid cells.
 - ▶ Discrete actions = N , E , W , S . Walls block motion.
 - ▶ Terminal states. The game ends in those cells.
- Noisy movement. For example, if the agent moves North:
 - ▶ 80% of the time, the agent goes North, if there is free space.
 - ▶ 10% of the time goes East and 10% goes West.
 - ▶ Walls block movement. Agent does not move.

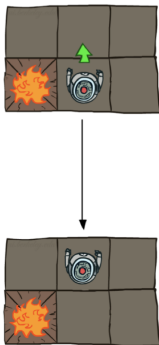


- Rewards at each step:
 - ▶ Small “living” rewards each step (positive or negative).
 - ▶ Large rewards at the end.

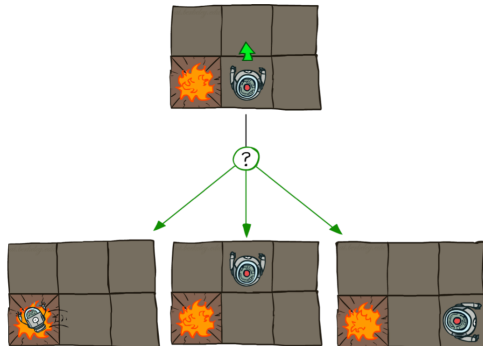
Credit: Dan Klein, Pieter Abbeel

Example. Gridworld

Deterministic motion

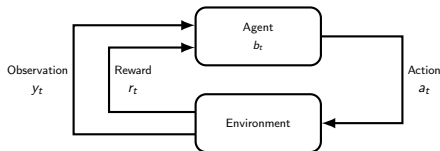


Stochastic motion



Credit: Dan Klein, Pieter Abbeel

Generalized Decision Processes (RL optative course)

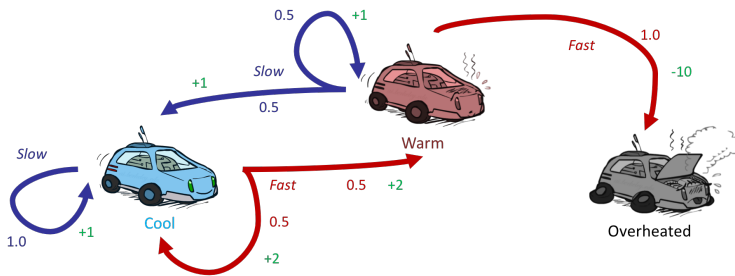


- The agent indirectly observes the environment. $y_t \sim p(y_t|x_t)$
- Partially Observable Markov Decision Processes (POMDPs):
 - ▶ State is *hidden*.
 - ▶ State can be estimated with beliefs/probabilities $b_t = p(x_t|y_{1:t}, a_{1:t})$.
 - ▶ Action $a_t \Leftarrow \pi(b_t)$
- Non-Markov Decision Processes:
 - ▶ Action $a_t \Leftarrow \pi(y_t)$
 - ▶ Policy much more complicated (e.g.: deep neural networks).
 - ▶ Maybe ill-posed.
- Reward can be stochastic $r_t \sim p(r_{t+1}|x_t, a_t)$

How is everything connected?

Models: environment predictions

- Transition model $p(x_{t+1}|x_t, a_t)$
- Observation model $p(y_t|x_t)$ ← We assume perfect observations $x_t = y_t$
- Reward model $p(r_{t+1}|x_t, a_t)$ ← We assume deterministic rewards $r_t = R(x_t, a_t, x_{t+1})$

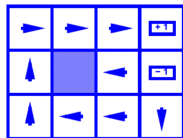


Credit: Dan Klein, Pieter Abbeel

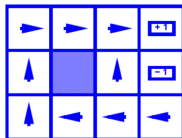
How is everything connected?

Policy: agent behavior

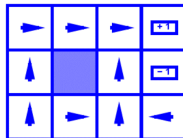
- Mapping from states x_t to actions a_t . We want the **optimal policy**.
- Deterministic $a_t = \pi(x_t)$ or stochastic $\pi(a_t|x_t) \Rightarrow a_t \sim p(a_t|x_t)$



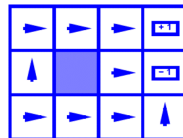
$R = -0.01$



$R = -0.03$



$R = -0.4$



$R = -2.0$

- What is the best policy? Why?

Credit: Dan Klein, Pieter Abbeel

How is everything connected?

Value function

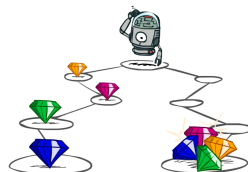
- Prediction of future rewards for a given policy.
- Informs about how good/bad is to reach a state.

$$V^{\pi}(x) = \mathbb{E}_{\pi} (r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots | x)$$

$$Q^{\pi}(x, a) = \mathbb{E}_{\pi} (r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots | x, a)$$

Discounting factor:

- Do you prefer 5\$ now or 10\$ in a week?
- $0 < \gamma \leq 1$
- Penalizes procrastination.

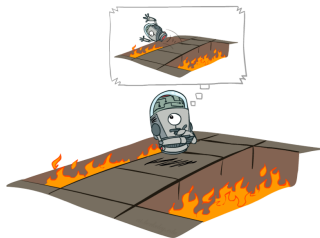


Credit: Dan Klein, Pieter Abbeel

Sequential decision making

Planning in MDPs (Offline)

- The agent has a good model of the environment.
- Everything is computed with the model. No real interaction.



Reinforcement learning (Online)

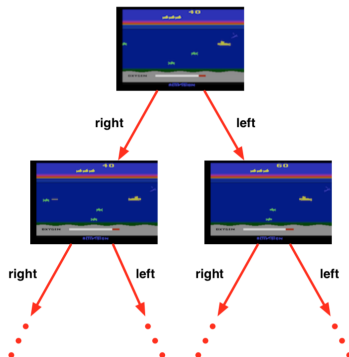
- The agent has minimal information of the environment.
- Learning by interaction.
- Exploration and exploitation



Credit: Dan Klein, Pieter Abbeel

Video game example: Planning

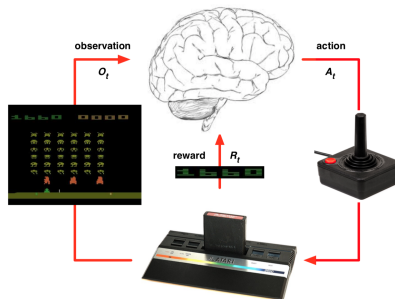
- Rules of the game are **known**
- We can emulate or simulate $\Rightarrow$ Access to perfect model.
- Given a state x_t and an action a_t :
 - We can predict the next state x_{t+1}
 - We can predict the reward of the step r_t .
- Find optimal policy by planning.
For example: tree search for discrete systems.



Credit: David Silver

Video game example: Reinforcement learning

- Rules of the game are **unknown**
- Learn the rules by interaction (playing)
- Move joystick (action) $\Rightarrow$ check pixels (state) and score (reward)
- Choose **good actions** to improve score (exploitation).
- Choose **new actions** to learn about game (exploration).



Credit: David Silver